THE LAWS OF SOFTWARE TEAMS

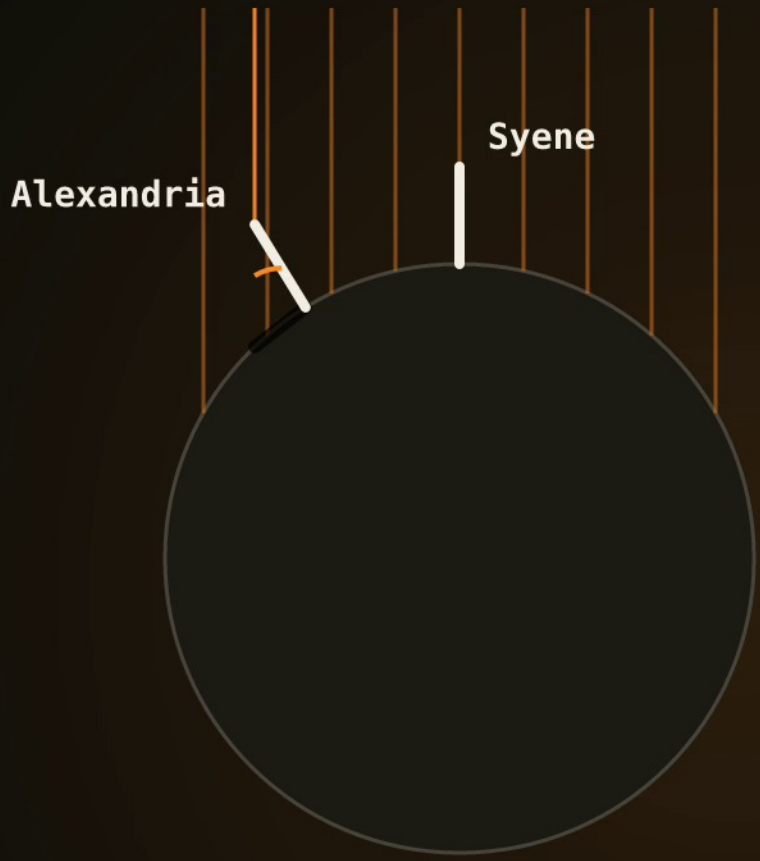
Gilb's law

Anything you need to quantify can be measured better than not at all

swipe →

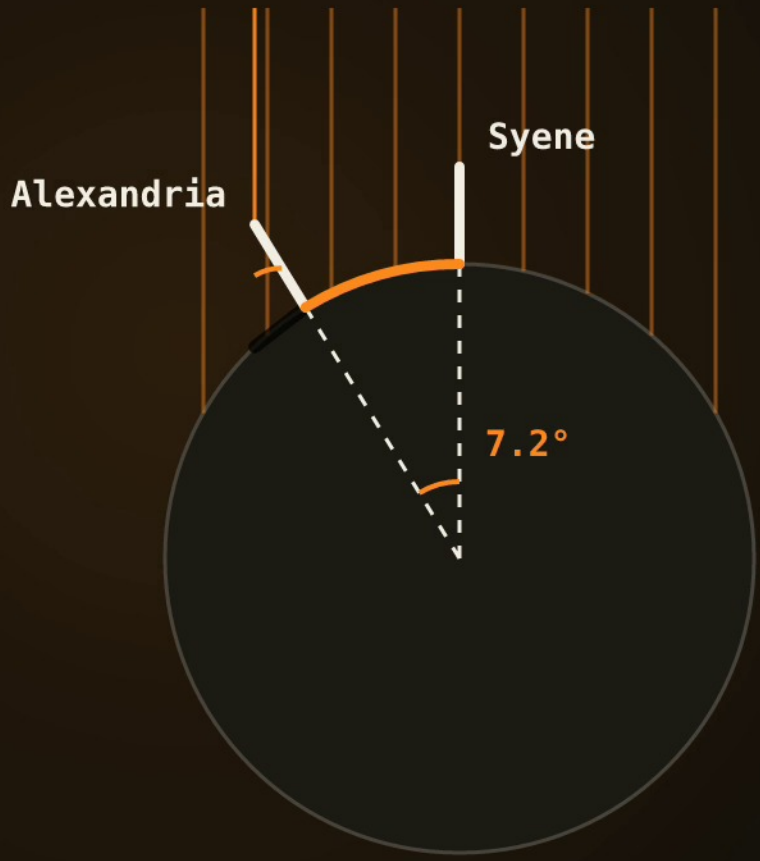
Ivan Rykov

He measured the Earth with a stick



~2,200 years ago. No satellites

Noon. No shadow in Syene
7.2° in Alexandria



7.2° = 1/50 of a circle

5,000 stadia × 50 = 250,000

Ivan Rykov · drawing not to scale

"Anything you need to quantify can be measured in some way that is superior to not measuring it at all"

Gilb's law

Tom Gilb · named in Peopleware, DeMarco & Lister



Measuring ≠ certainty
It's less uncertainty

How long until a new hire ships?

weeks · example



first guess: 1-20 weeks

after 5 answers: 4-7 weeks

Douglas W. Hubbard, How to Measure Anything

You need less data than you think



Pick 5 at random

94% chance the median is inside their range

Hubbard's Rule of Five · $1 - 2 \times (½)^5 = 93.75\%$

Two laws, one balance

GOODHART

Don't make it a target

GILB

Don't stop measuring

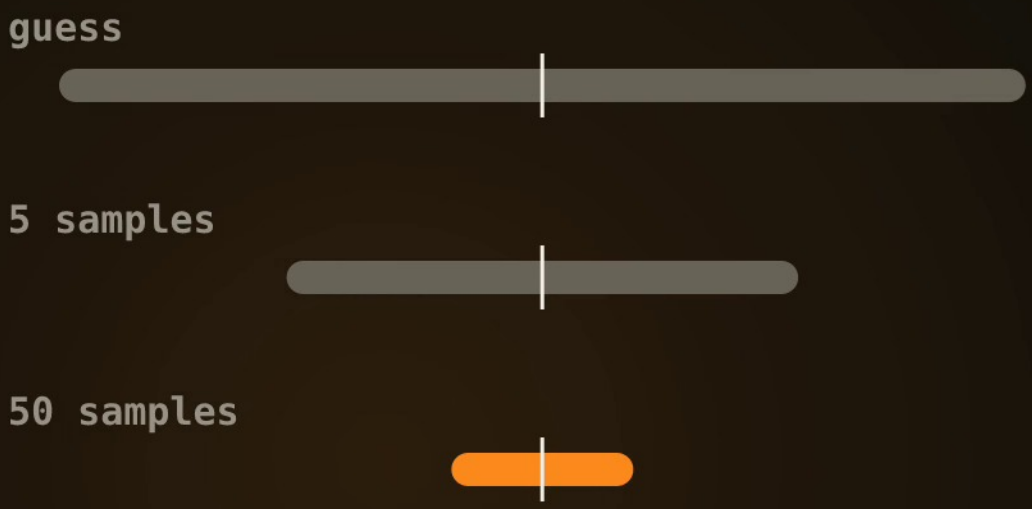
Flying blind is worse

Start small

1. **Name the decision**
   the number should change
2. **Guess a range**
   e.g. 2-12 weeks
3. **Sample a little**
   and watch the range shrink

Hubbard: if it can't change a decision,
it isn't worth measuring

A rough number beats a confident guess



TAKEAWAY

A rough number beats a confident guess

Sources: Tom Gilb, via Peopleware · Douglas Hubbard, How to Measure Anything · Eratosthenes

next: Goodhart's law